

**G. PULLA REDDY ENGINEERING COLLEGE (Autonomous): KURNOOL**  
**Department of Emerging Technologies in Computer Science**

**TITLE: Smart Loan Recovery System Using Machine Learning**

**Priority – 3**

**Abstract:**

Loan defaults represent a critical challenge for financial institutions, significantly impacting their profitability and cash flow stability. This project focuses on the development of a Smart Loan Recovery System leveraging machine learning to enhance the efficiency of loan collection processes. Using historical data that includes borrower demographics, loan attributes, repayment patterns, and past collection efforts, the system is trained to predict loan recovery outcomes. The dataset features vital indicators such as age, employment status, income, loan amount, interest rate, EMI history, and methods of recovery applied. By analyzing these variables, the model can classify loans as fully recovered, partially recovered, or unrecovered. The system aids institutions in identifying high-risk borrowers, prioritizing recovery strategies, and minimizing collection costs. Machine learning algorithms are employed to uncover hidden patterns in repayment behavior, enabling more accurate forecasting and optimized decision-making. This approach not only enhances recovery rates but also contributes to more sustainable lending practices in the financial sector.

**Project Domain:**

Machine Learning, Predictive Analytics

**Software and Hardware Requirements:**

- Software: Python, Scikit-learn, Pandas, Matplotlib, Jupyter Notebook
- Hardware: System with at least 8GB RAM, optional GPU for model training
- Tools: Google Colab or local IDE for development

**Student Details:**

| S.No. | Roll No    | Name of the Student | Signature |
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